

Exploratory Data Analysis of RIT Opportunity Wise Dataset

By

RIT 0806 AI Data Team 17

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The team worked together to clean and prepare the RIT Opportunity Wise Dataset, ensuring its correctness, consistency, and readiness for research. This method entailed detecting and rectifying data abnormalities, deleting duplicate entries, restructuring and standardizing columns, and addressing data quality concerns. These initiatives strengthened the dataset's general reliability and integrity, laying the groundwork for precise analysis, prediction, visualization, and reporting.

1. Introduction

Educational and professional development activities are critical for students to learn practical skills, gain industry experience, and improve their career preparation. Understanding how students use these opportunities can provide important insights into participation patterns, application outcomes, and success determinants. The RIT Opportunity Wise Dataset comprises records of learner applications for a variety of opportunities, including demographics, academic backgrounds, country of residence, opportunity categories, and application status. By studying this dataset, institutions can gain a greater understanding of student engagement, discover patterns in participation, and assess the efficiency of various programs. The findings of this analysis can help with data-driven decision-making targeted at improving student experiences, increasing participation, and improving overall program outcomes.

2. Executive Summary

The RIT Opportunity Wise Dataset contains 8,558 application records spanning 23 opportunities across five categories: internships, courses, events, competitions, and engagements, collected between January 2023 and March 2024. Analysis shows that internships dominate the dataset (63.3%) and have a high rejection rate of 65.8%, highlighting the competitive nature of these opportunities. The largest group of applicants comes from the United States, while Information Systems and Computer Science are the most represented academic majors. Data quality issues were identified, including invalid Apply Date timestamps and missing Opportunity Start Date values, which should be resolved before deeper analysis. Overall, the dataset reflects the broad international reach of RIT opportunities while revealing areas for further investigation, particularly regarding engagement patterns and fairness in selection outcomes.

3. Statement of Problem

Despite having extensive data on students and opportunities, there is limited insight into the factors that influence student participation and application outcomes within RIT programs. Key patterns related to demographic engagement, opportunity preferences, application success, and disparities across countries, genders, and academic majors remain unclear. Additionally, important metrics such as application volume, rejection rates, team allocations, awards, and dropout rates have not been fully explored. This lack of understanding may hinder efforts to improve program

accessibility, engagement, and student success. Therefore, this project seeks to analyze the RIT Opportunity Wise Dataset to uncover meaningful trends, assess learner outcomes, and provide data-driven recommendations for program improvement and decision-making.

4.1. Dataset Overview

The RIT Opportunity Wise Dataset is a structured tabular record of student applications to various opportunities hosted through RIT's engagement platform.

Attribute	Value
Total Records (Rows)	8,558
Total Features (Columns)	16
Unique Opportunities	23
Opportunity Categories	Internship, Course, Event, Competition, Engagement
Date Range	January 2023 – March 2024
Unique Academic Institutions	2,089
Missing Values	Institution Name, Opportunity Start Date, and Current/Intended Major.

4.2. Types of Opportunities Included

The dataset includes five separate opportunity categories, each with a different developmental aim for students:

Category	Description	No. of Records
Internship	Structured professional work experiences with industry partners	5,421
Course	Skill-building academic or professional development courses	2,037
Event	Workshops, seminars, networking, and informational sessions	545
Competition	Academic or professional challenge competitions	425
Engagement	Community or institutional engagement activities	130

4.3. How Student Participation and Engagement Are Recorded

Each entry in the collection represents a student's engagement using a combination of temporal markers and status attributes. Student participation is primarily measured through:

- **Opportunity ID:** This column stores the unique ID of the opportunity.
- **Opportunity Name:** This column stores the name of the opportunity.
- **Opportunity Category:** This field tells us the type of opportunity, such as an internship, course, competition, or event.
- **Opportunity Start Date:** This field shows the start date of the opportunity.
- **Opportunity End Date:** This field shows the end date of the opportunity.

5.1. Data Quality Report

- Data quality issues were identified in dates and timelines, including 307 malformed application dates, 43 cases where applications occurred before sign-up, and 527 records with opportunity end dates occurring before start dates.
- Record integrity issues were found, including 237 duplicate applications, 18 test/placeholder records, 22 entries containing IDs instead of names, 14 single-character names, and 1 unrealistic age record.
- Data completeness and consistency concerns remain, with 3,794 missing Opportunity Start Dates (about 44% of the dataset), a small number of missing institution and major values, and inconsistent country name formats that may affect analysis accuracy.

5.2. Data Cleaning

- Data consistency was improved by standardizing date formats and correcting capitalization in institution names, countries, and academic majors.
- Missing values and formatting issues were addressed, with 3,804 blank cells (mainly in the Opportunity Start Date column) cleaned, resulting in a dataset with no missing values.

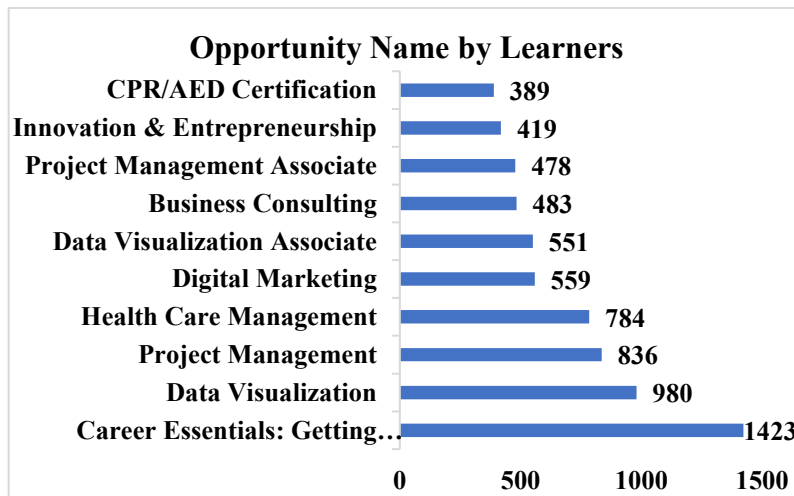
Data Dictionary

The following spreadsheet has a comprehensive data dictionary. Data Dictionary (Spreadsheet Form) [link](#).

Exploratory Data Analysis (EDA) & Visualizations

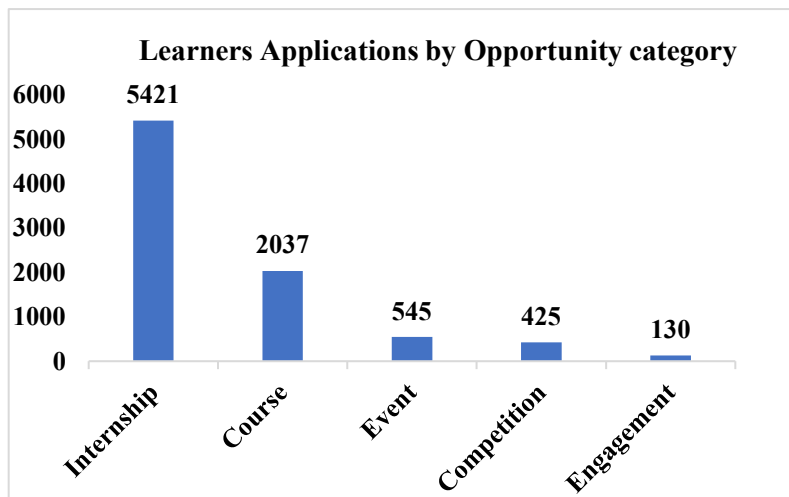
Key Findings

1. Top Opportunities by Number of Learners.



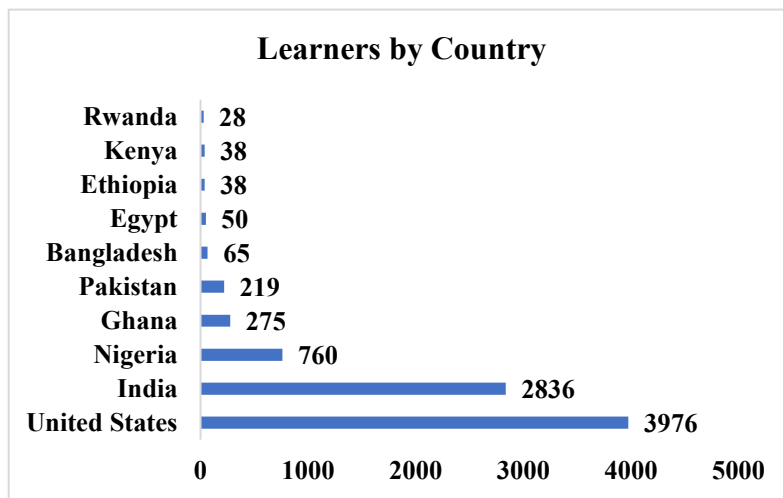
Career Essentials drew the most learners, exceeding all other options.

2. Applications by Opportunity Category.



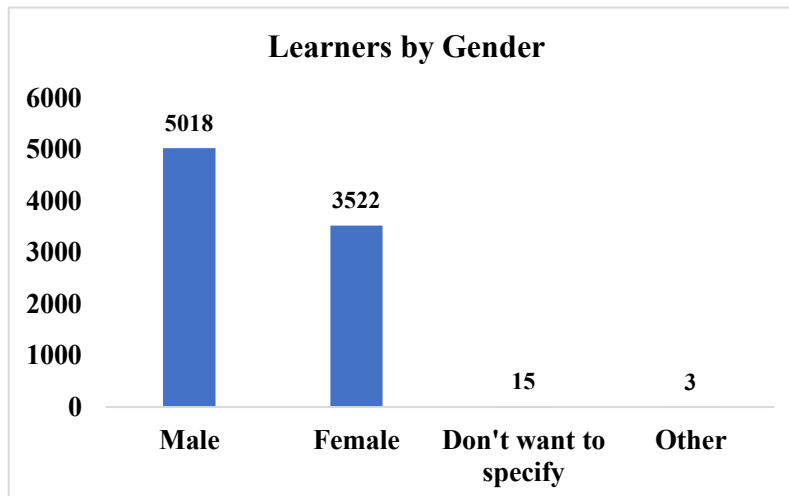
Most applications were submitted for internships, accounting for the largest share of total applications.

3. Top Countries by Learner Participation.



The United States had the highest number of learners, followed by India and Nigeria. Participation from other nations was significantly lower, indicating a concentration of learners in a few places.

4. Distribution of Learners by Gender.



Male students made up the majority of participants, with female students accounting for a sizable minority. Only a small percentage of students chose "Other" or preferred not to indicate their gender.

Notable Trends, Patterns, or Relationships in the Data

- **Participation is concentrated in a few countries:** the majority of learners come from the United States, India, and Nigeria, with participation from other countries being low, indicating a limited global reach.
- **Gender representation remains uneven:** Male learners regularly make up the majority of participants, indicating a potential gender diversity gap in program engagement.
- **Internships are the most sought-after opportunities:** Internship programs receive much more applications than courses, events, competitions, or engagements, showing students' strong preference for career-oriented and hands-on experience opportunities.

Recommendations

- **Broaden global participation:** To establish a more varied and globally representative learning community, expand outreach efforts to underrepresented regions such as Africa, Southeast Asia, and Latin America.
- **Promote gender diversity:** Implement targeted recruitment strategies, such as collaborations with women-focused groups and inclusive marketing campaigns, to boost female engagement and diversity.
- **Expand internship offerings:** Increase the amount of internship possibilities and look into hybrid internship-learning programs to better match learner demand and boost application success rates.

Tools Used

Ms. Excel

Python